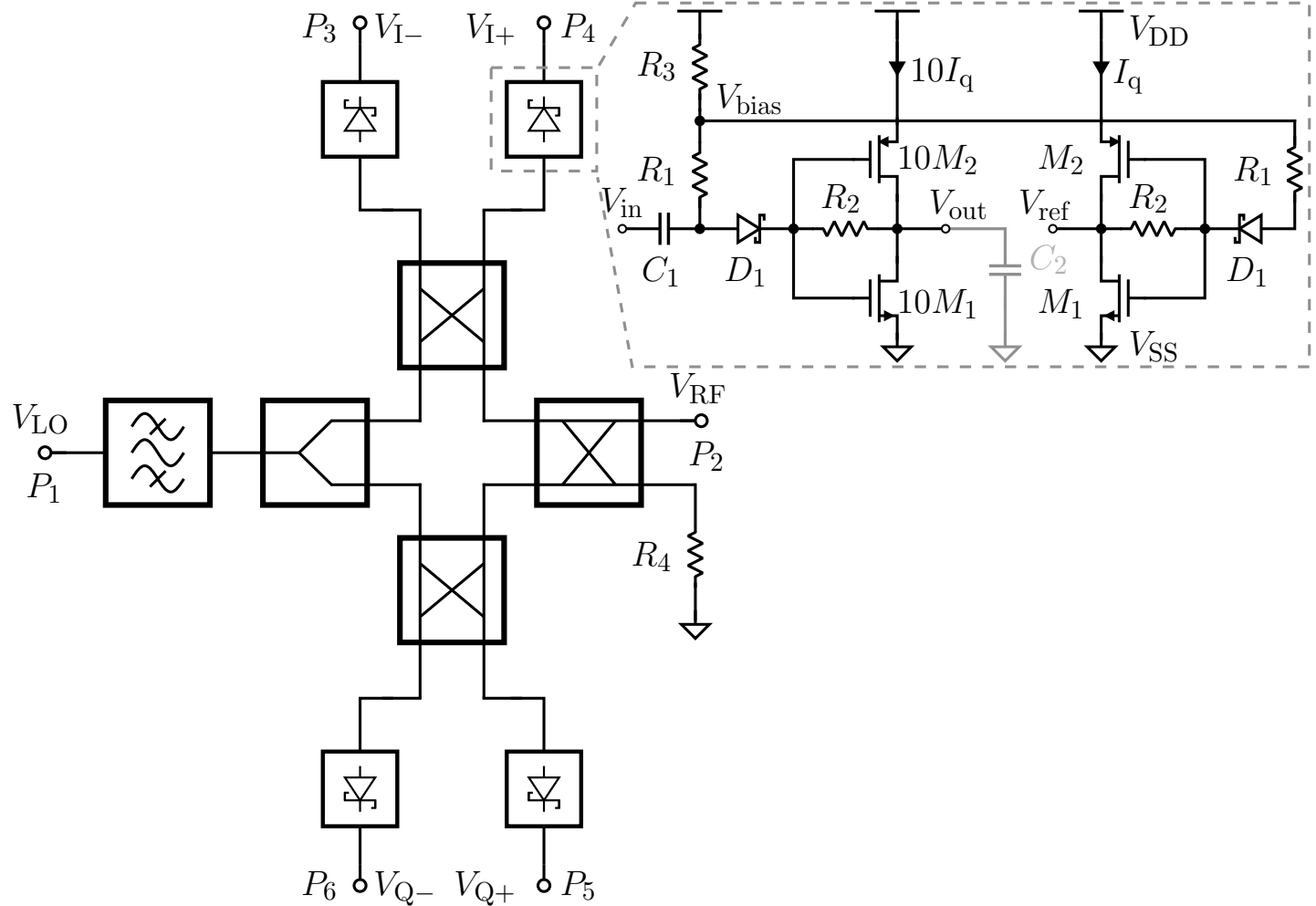

P_3
$V_{\mathrm{I}-}$
$V_{\mathrm{I}+}$
P_4
V_{LO}
P_1
V_{RF}
P_2
R_4
P_6
$V_{\mathrm{Q}-}$
$V_{\mathrm{Q}+}$
P_5
R_3
V_{bias}
R_1
V_{in}
C_1
D_1
$10I_{\mathrm{q}}$
$10M_2$
R_2
V_{out}
$10M_1$
C_2
V_{DD}
I_{q}
M_2
V_{ref}
M_1
V_{SS}